

Reorder Buffer (ROB) — Internal Architecture

Client Interface (CIF) — Read Return Path

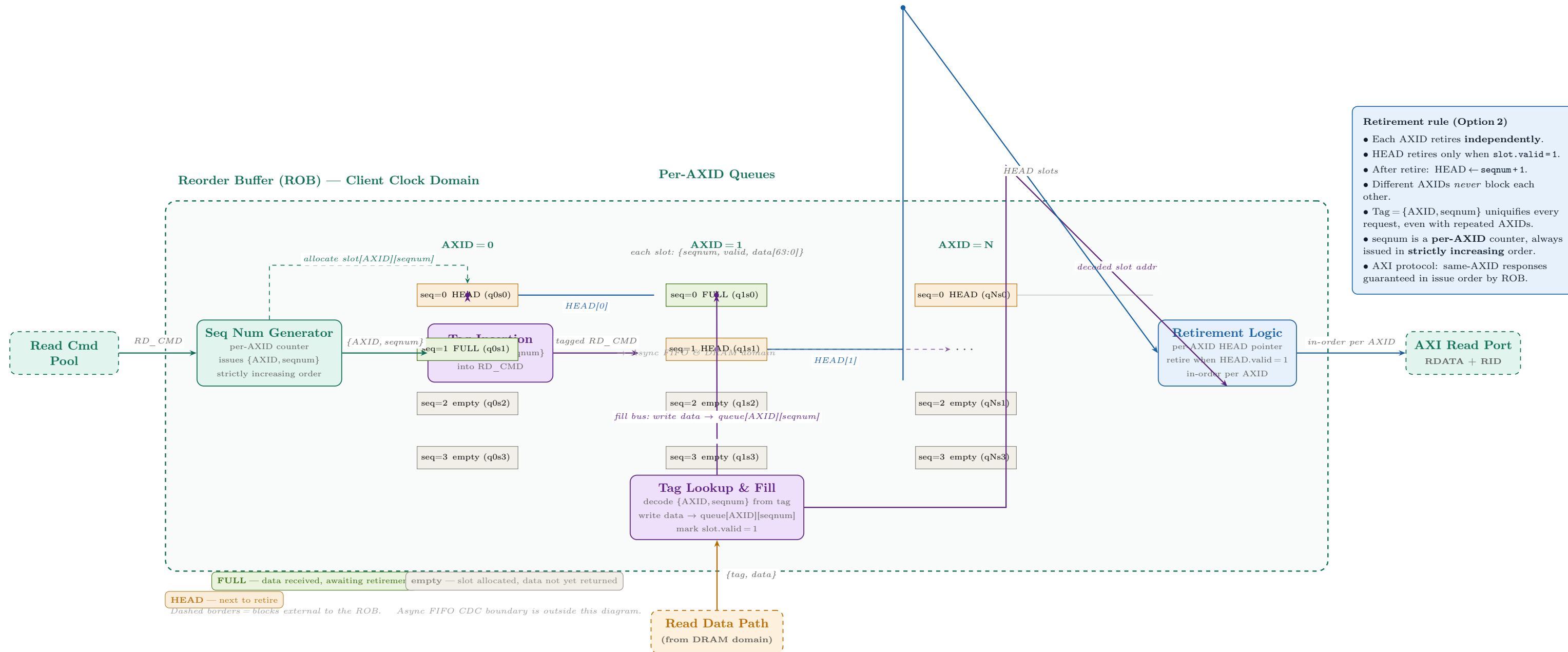


Figure 1: Internal architecture of the Reorder Buffer (ROB), operating entirely within the client clock domain. **Issue path:** the **Seq Num Generator** assigns a per-AXID **seqnum** in strictly increasing order, allocates the corresponding **queue[AXID][seqnum]** slot, and passes $\{AXID, seqnum\}$ to **Tag Insertion**, which embeds the composite tag into the outgoing **RD_CMD** before it crosses the Async FIFO CDC boundary (outside ROB). **Return path:** read data arrives from the DRAM domain as $\{tag, data\}$; **Tag Lookup & Fill** decodes the tag, writes data into **queue[AXID][seqnum]**, and sets **slot.valid = 1**. **Retirement Logic** advances a per-AXID HEAD pointer whenever **HEAD.valid = 1**, guaranteeing in-order delivery per AXID (Option 2) while allowing independent AXIDs to retire without mutual blocking.